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Species

Species is the basic unit of classification in the taxonomic hierarchial system. It is a group of animals having similar morphological features (traits) and is reproductively isolated to produce fertile offspring. There are some exceptional animals which can produce **sterile offspring** because of mating with closely related species (Figure 1.2).

Crosses between

Male horse and Female Donkey results in Hinny (Sterile).

Male Donkey and Female Horse results in Mule (Sterile)

Male Lion and Female Tiger results in Liger

Male Tiger and Female Lion results in Tigon

Genus: It is a group of closely related species which have evolved from a common ancestor. In some genus there is only one species which is called as **monotypic genus** (e.g. Red panda is the only species in the genus *Ailurus* : *Ailurus fulgens*) (Figure 1.3). If there are more than one species in the genus it is known as **polytypic genus,** for example 'cats' come under the Genus *Felis*, which has a number of closely related species, *Felis domestica* (domestic cat), *Felis margarita* (jungle cat). *Felis silvestris* (wild cat)



Figure 1.3 Red Panda – *Ailurus fulgens*

Systematics of Human being

KINGDOM
Animalia

PHYLUM
Chordata – Animals with a notochord or vertebral column (Back bone)

CLASS
Mammalia – Body covered with hair or fur. Mammary glands are present

ORDER
Primata – Mammals with forward looking eyes and grasping fingers

FAMILY
Hominidae – Primates with relatively flat faces and binocular vision

GENUS
Homo – Hominids with large brain and upright posture

SPECIES
Homo sapiens – Bipedal and higher intelligence